Architecting Data Ingestion and Data Exploitation Solution Patterns

Part II



Objectives and Scope

This section focuses on the target architecture to 1) ingest batch datasets from data.gov.sg, into an AWS S3 data lake inside HDB's private VPC, and 2) allowing the Data Science Team to query from the lake using Tableau running on AWS as a separate VPC, keeping traffic private.

Design Pillars

- AWS Native
- Secure
- Reliable
- Scalable
- Maintainable
- Performant

Presentation and Deliverables

I will be presenting the requirements as a separate section, solutioning and explaining the ingestion and exploitation parts as a distinct part to ensure the diagram does not become too small.



Assumptions

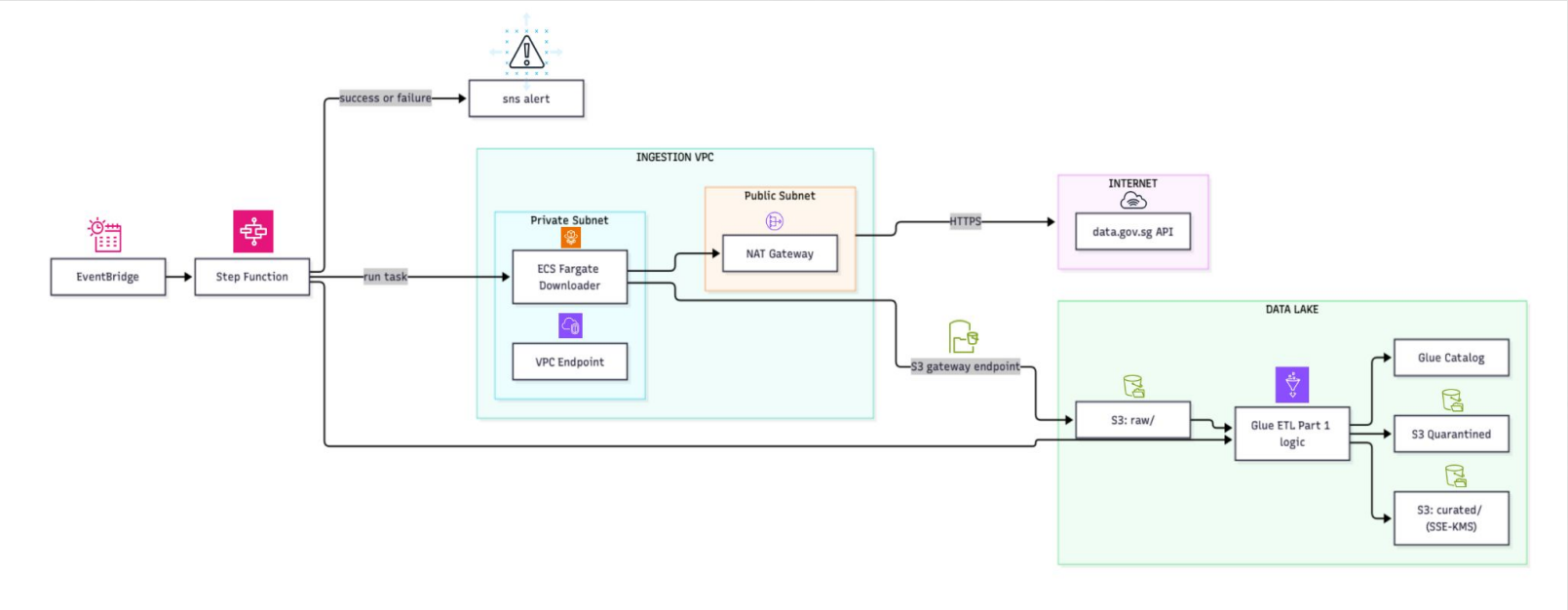
This section highlights some considerations and assumptions taken in the design on the architecture.

1. Three workload accounts are used: Ingestion, Data Lake, and Analytics. If HDB uses a single account, the same design applies with separate VPCs and IAM rules
2. data.gov.sg is reached via HTTPS through the public API
3. Once public data.gov.sg information is combined with HDB's internal data, the output is treated as restricted
4. Tableau Server runs on EC2 in Analytics VPC
5. Pipeline logic in part 1 is packaged as glue jobs

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Data Ingestion

Ingestion Architecture



Ingestion

Flow of events

Step 1	EventBridge Scheduler starts a Step Function state machine on cron
Step 2	Step Function reads config file in S3 and checks data.gov.sg
Step 3	For each changed dataset, step function runs ECS Fargate task in private subnet
Step 4	Task gets API secret from secret manager and downloads file through API (nat gateway)
Step 5	Downloaded files land in S3
Step 6	Glue Job runs part 1 transformation and validation as glue job. Successful validation lands parquet files in curated/, validation failures land in Quarantined/
Step 7	Glue catalog table is updated

Ingestion - Large Files (100Mb)

Large File Ingestion Considerations

Consideration 1	Fargate v Lambda	Fargate is used as it supports up to 200G of scratch space, and does not have 15min timeout limit like lambda
Consideration 2	AWS Glue	Glue handles heavy transformation, joins, and format conversions to keep things scalable.
Consideration 3	Stream Multipart Upload	Instead of loading whole file into memory, pipeline streams data into S3 using parallel multipart upload which has checksum verification, replay, and high reliability during network outages
Consideration 4	Idempotency	Ingestion script to use exponential backoff retries. Each run writes unique run_id path, rerunning failed jobs never overwrite existing data



Ingestion - Network Segmentation

Network Segmentation Considerations

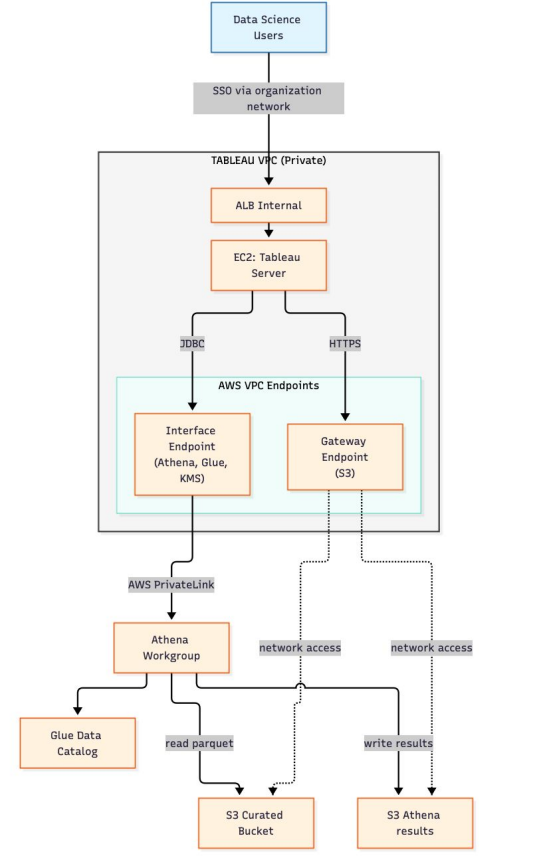
Consideration 1	Private Subnet	All Fargate tasks to execute in private subnet with security groups locked down to egress only.
Consideration 2	NAT Gateway	NAT Gateway placed in a public subnet to allow outgoing request to data.gov.sg API. No inbound routes from public network to any compute instances
Consideration 3	Lake access through VPC endpoints	S3 Gatewat Endpoint allows fargate and glue to talk to S3 within AWS by securing each endpoint policy to HDB's specific S3 buckets



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Data Exploitation

Exploitation Architecture



Step 1	DS Users sign in to Tableau server SSO through internal load balancer
Step 2	Tableau connects to athena JDBC connection through EC2 role
Step 3	Request goes to Athena via VPC endpoint
Step 4	Athena queries Glue Data Catalog to locate table schemas the reads parquet files from S3 through S3 gateway endpoint
Step 5	Athena write query results to results bucket and returns it to Tableau. For heavy extraction, to use scheduled Tableau extracts to pre-load and cache data to reduce query load on Athena

Exploitation - Private Traffic

Private Traffic Considerations

Consideration 1	No internet exposure	Tableau VPC has no internet gateway. All AWS services to be accessed via private interface endpoints and S3 through gateway endpoint.
Consideration 2	No network level connections	Tableau VPC - Data Lake VPC is blocked at the network layer. Communication is managed via AWS service endpoints controlled by IAM and endpoint policies to curb vulnerable network level connections
Consideration 3	S3 bucket policy	S3 bucket to accept traffic only from whitelisted VPC endpoints



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